Aug 11 Conditional probability

A, B : two events

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

B: event that occured "additional info"

Multiplication rule

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) P(A)$$

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) P(A_2|A_1) P(A_3|A_1 \cap A_2) \cdots P(A_n | A_1 \cap A_2 \cap \cdots \cap A_{n-1})$$

$$P(A_1 \cap A_2 \cap \cdots \cap A_n | B) = P(A_1|B) P(A_2|A_1 \cap B) P(A_3|A_1 \cap A_2 \cap B) \cdots P(A_n | A_1 \cap A_2 \cap \cdots \cap A_{n-1} \cap B)$$

$$P(A|A) = 1$$

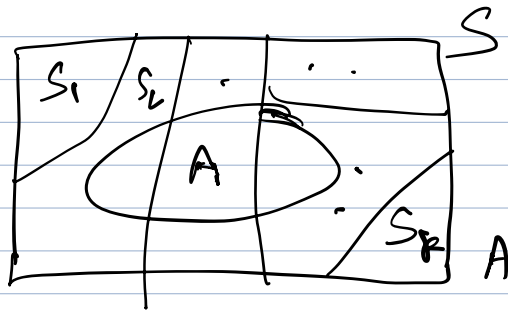
Ex: Birthday problem

n

$$P(\text{no two have same birthday}) = 1 \cdot \left(1 - \frac{1}{365}\right)\left(1 - \frac{2}{365}\right) \cdots \left(1 - \frac{n-1}{365}\right)$$

Partitions

$$P(A) = \sum_{i=1}^{k} P(A \cap S_i)$$

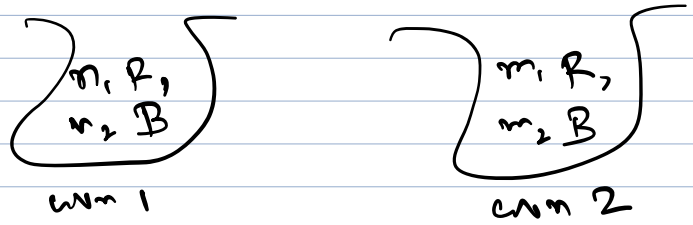


$$S = S_1 \sqcup S_2 \sqcup \cdots \sqcup S_k$$

↑ disjoint union

$$A = (A \cap S_1) \sqcup (A \cap S_2) \sqcup \cdots \sqcup (A \cap S_k)$$

Ex:



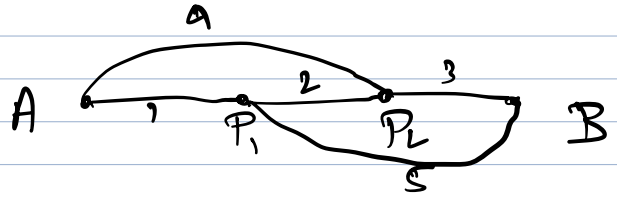
Experiment: Pick an urn at random.
Pick a marble at random from the chosen urn

A: marble is red

$$P(A) = P(A \mid \text{urn 1}) \cdot P(\text{urn 1}) + P(A \mid \text{urn 2}) \cdot P(\text{urn 2})$$

$$= \frac{n_1}{n_1+n_2} \cdot \frac{1}{2} + \frac{m_1}{m_1+m_2} \cdot \frac{1}{2}$$

Ex:



Monty Hall problem

Game show. 3 doors. 1 door: car behind
2 doors: goat behind

You say Door 1. Host opens Door 3 & there is a goat behind it.

Switch (or) Stay?

→ all 6 orders are equally likely

$G_1 \longleftrightarrow G_2 \longleftrightarrow C$

D1	D2	D3

→ host will show you a goat.

1	2	3	your pick	host	
G_1	G_2	C	1	2	$\frac{1}{6} \times \frac{1}{3}$
"	"	"	2	1	"
"	"	"	3	1	$\frac{1}{6} \times \frac{1}{3}$

(G_2, G_1, C)

⋮

Ex: Family has 2 children.

(a) At least one is a boy. What is the prob. that the family has 2 boys?

(b) You meet one child & see that it is a boy. What is the prob. of 2 boys?